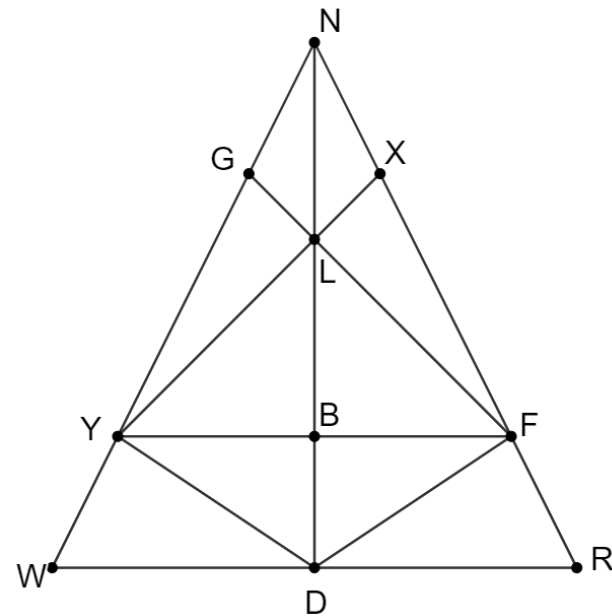


1. Triangle NRW is shown where $\overline{LB} \perp \overline{YF}$ and $\overline{DB} \perp \overline{WR}$.

Given: $\overline{LB} \perp \overline{YF}$, $\overline{DB} \perp \overline{WR}$

Prove: $\triangle YBN \sim \triangle WDN$



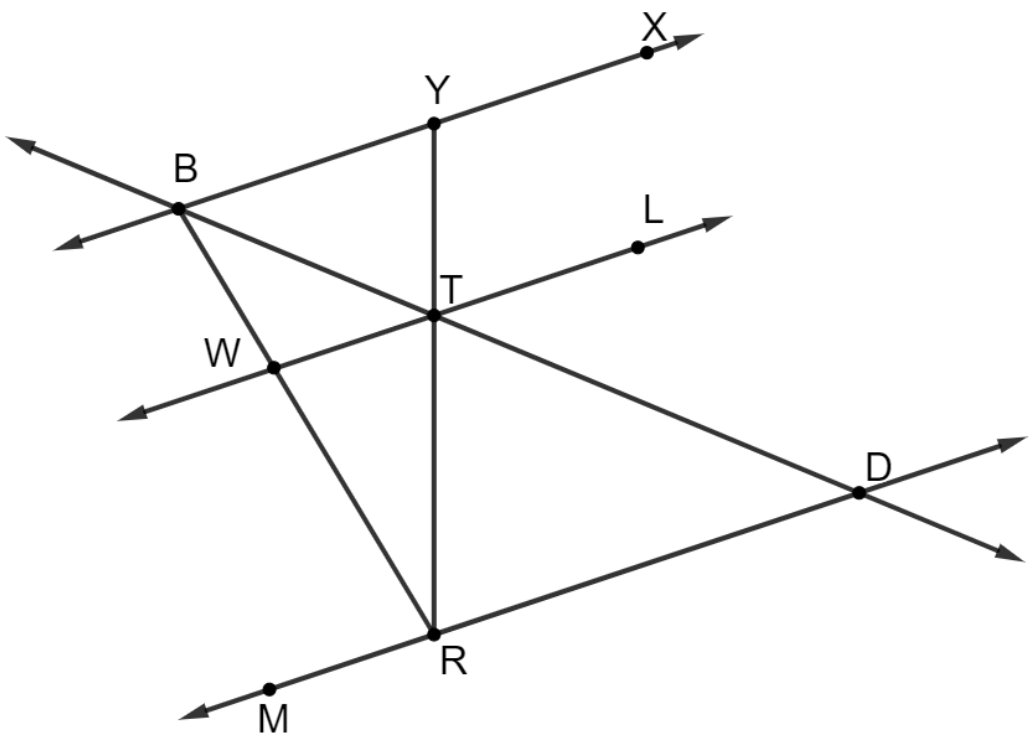
An incomplete proof is shown.

Statement	Reason
1. $\overline{LB} \perp \overline{YF}$, $\overline{DB} \perp \overline{WR}$	1. Given
2. $\angle YBN$ is a right angle; $\angle WDN$ is a right angle	2.
3. $m\angle YBN = 90^\circ$ and $m\angle WDN = 90^\circ$	3. Definition of a right angle
4. $m\angle YBN = m\angle WDN$	4. Transitive property of equality
5.	5. Reflexive Property
6. $\triangle YBN \sim \triangle WDN$	6.

What is the reason for step 6?

Reason for step 6	
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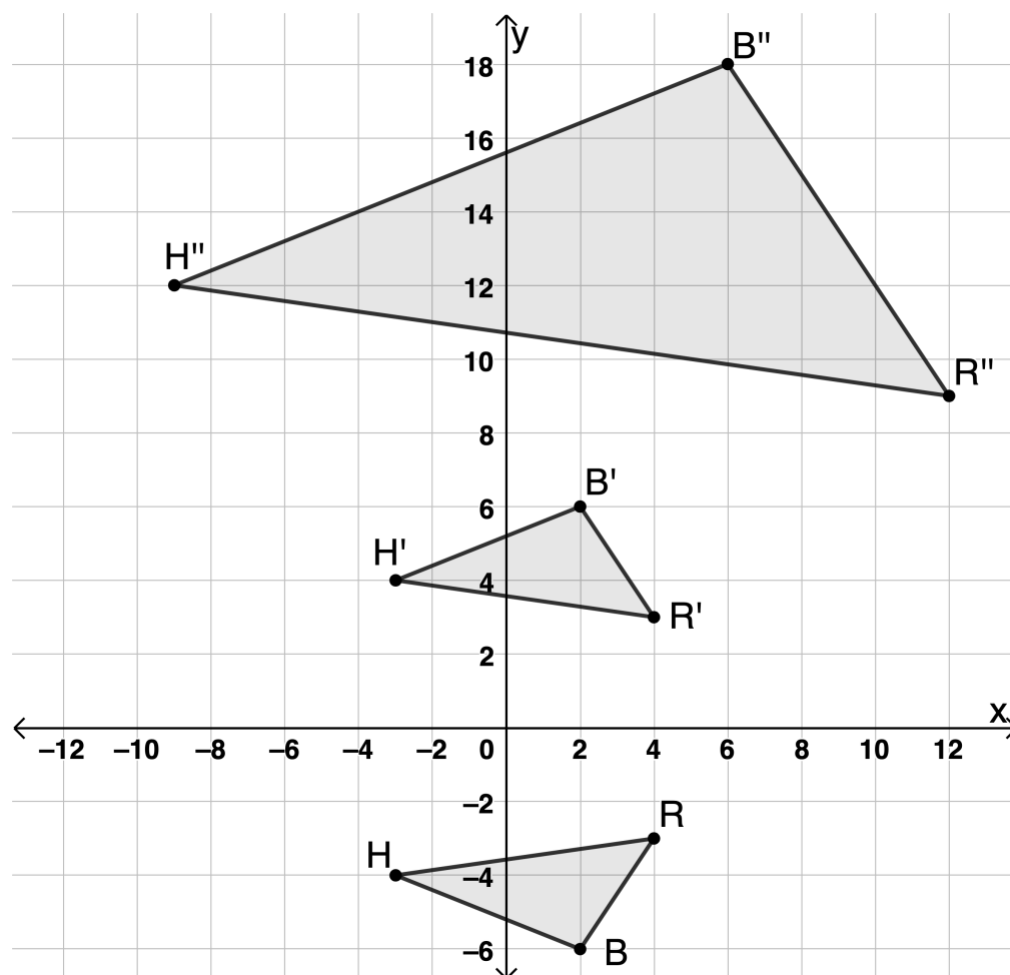
2. A diagram is shown.



For each set of given information, complete the missing information in the table.

	Given Information	Triangles	Are They Similar?	Theorem or Postulate Used
a.	$\frac{BT}{BD} = \frac{BW}{BR}$	$\triangle DBR$ and $\triangle TBW$	☐ Yes ☐ No	
b.	$\overrightarrow{BX} \parallel \overrightarrow{WL} \parallel \overrightarrow{MD}$ and $\frac{BW}{WR}$	$\triangle TWB$ and $\triangle TWR$	☐ Yes ☐ No	
c.	$\overrightarrow{BX} \parallel \overrightarrow{MD}$	$\triangle YTB$ and $\triangle RTD$	☐ Yes ☐ No	
d.	$\frac{BR}{WR} = \frac{YR}{TR} = \frac{YB}{TW}$	$\triangle YRB$ and $\triangle TRW$	☐ Yes ☐ No	

3. Triangles HRB , $H'R'B'$, and $H''R''B''$ are shown on the coordinate grid.

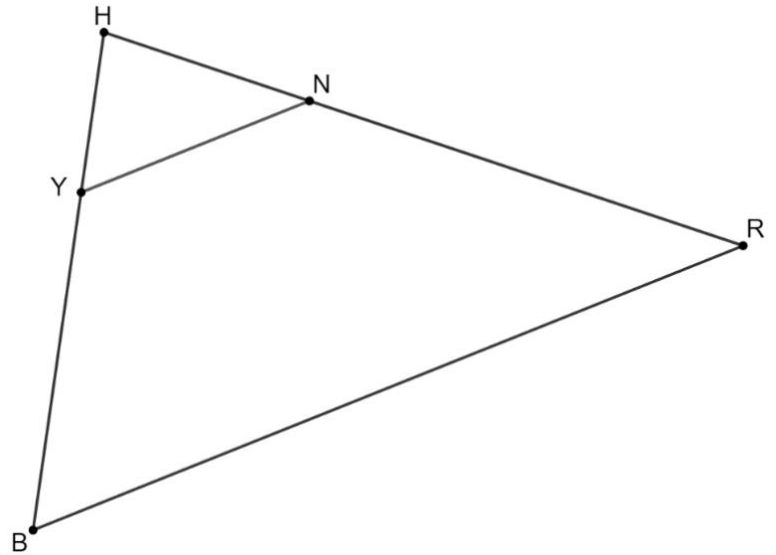


- a. Complete the table using algebraic descriptions so that the sequence of transformations will map $\triangle HRB$ onto $\triangle H''R''B''$.

Sequence	Algebraic Description
Transformation 1	
Transformation 2	

- b. Use the definition of similarity in terms of non-rigid transformations to justify that $\triangle HRB \sim \triangle H''R''B''$.

5. A paleontologist is setting up boundaries to dig for fossils. $\triangle HYN$ represents the area where invertebrate fossils have already been found. The paleontologist extends the dig area by drawing $\overline{BR}$ parallel to $\overline{YN}$ so that $\frac{YN}{BR} = 0.32$.



The paleontologist uses the diagram and the given information to extend the dig site. All measurements are in meters (m).

$HY = (6x + 4)$ m, $HN = (11x - 19.5)$ m, $NR = (18.5x)$ m, and $YN = 85$ m.

- Determine the distance, in meters, between points B and R . Show your work.
- Determine the value of x . Show your work.
- Determine the distance, in meters, between points B and Y . Show your work.